COURSE SLIDES

# Microsoft 365 Copilot for Excel

Course Code: C34

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

# Welcome & Housekeeping


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TRAINERS LISTED FOR C34

# Course Trainer Profiles

1

## Ken Hiong

Excel practitioner and adult educator with experience in business analysis, planning and reporting.

2

## Jim Gan Chiu Liang

Data automation specialist with extensive experience building practical Excel workflows.

3

## Liew Sing Loon

Technology and management educator with applied quantitative-analysis experience.

4

## Dwight Nuwan Fonseka

Data science leader and adult educator specialising in analytics, machine learning and responsible AI.


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ICE-BREAKER

# Let's Know Each Other

1. Your name, organisation and role.
2. One HR, recruitment or people report you build in Excel today.
3. One people decision your data should help someone make faster.
4. Whether you can see the Copilot button in Excel, and on which app (Windows, Mac or web).


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HOUSEKEEPING

# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.


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COURSE MATERIALS

# Access Course Materials

1. Open the C34 course page or the trainer-supplied Drive folder
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the Activities folder for all 8 lab folders
5. Keep the files together as you work

**Course page: tertiarycourses.com.sg/microsoft-365-copilot-for-excel.html**


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SCHEDULE

# Lesson Plan — 1 Day, 7.5 instructional hours

## Course sequence

- Day 1 — Analyse, visualise, dashboard and recommend with Copilot in Excel on HR and recruitment data
  - Topic 1: Use Copilot to Analyse Data (Labs 1–2)
  - Topic 2: Use Copilot to Plot Charts (Labs 3–4)
  - Topic 3: Use Copilot to Create Dashboards (Labs 5–6)
  - Topic 4: Use Copilot to Draw Insights and Recommendations (Labs 7–8)

## Day format

- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
  - 7.5 hours · 8 connected labs
- Learning arc
  - Concept → demonstrate → build → verify → discuss → recap


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WHAT YOU'LL ACHIEVE

# Learning Outcomes

1

## Analyse Data

Explain how Copilot in Excel works (modes, requirements and limits) and use it to analyse, highlight, sort, filter and calculate HR and recruitment data with verified formulas.

2

## Plot Charts

Use Copilot to create, refine and verify charts, PivotCharts and Python visuals that answer specific HR and recruitment questions.

3

## Create Dashboards

Use Copilot to build an interactive, refreshable HR KPI dashboard with PivotTables, KPI cards, charts and slicers, and check every KPI against its definition.

4

## Insights & Recommendations

Use Copilot and Python-based advanced analysis to draw evidence-backed insights and actionable, owned recommendations from workforce and survey data.


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LEARNING APPROACH

# How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes


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CORE CONCEPTS

# Copilot in Excel for HR and Recruitment


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OVERVIEW

# The BrightPath Scenario

1

## BrightPath Talent Solutions

A fictional Singapore recruitment, flexible-staffing and HR-outsourcing firm.

2

## Recruitment data

Candidate pipeline, job requisitions, monthly funnel metrics and contract placements.

3

## Workforce data

Internal headcount, exits, attrition drivers and an engagement pulse survey.

4

## Your role

HR / talent analyst turning these workbooks into verified answers, charts, dashboards and actions.


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THE BUILDING BLOCKS

# G-C-D-C-O-R Prompt Framework

1

## Goal

One task and the HR decision or workbook change it supports.

2

## Context

Audience, business situation, units and time period.

3

## Data

Named table, sheet, columns, grain and filters (Microsoft: Source).

4

## Constraints

What must not change, privacy limits and exclusions.

5

## Output

Number, formula column, PivotTable, chart, dashboard or summary.

6

## Review

Logic, assumptions, edge cases and the check you will run (Microsoft: Expectations).


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C34 OPERATING PRINCIPLE

# Copilot accelerates the work; your verification makes it trustworthy.

Describe the task precisely, inspect what Copilot built, and reconcile every number before it reaches a people decision.


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COURSE ROADMAP

# Four Topics, Eight Labs

## Analyse & Plot

- Labs 1-2: question, highlight, filter, formulas
- Labs 3-4: charts, PivotCharts, Python visuals

## Dashboard

- Lab 5: recruitment KPI dashboard
- Lab 6: workforce and attrition dashboard

## Insight & Recommend

- Lab 7: attrition drivers with Python
- Lab 8: survey themes to recommendations


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# The Learning Arc in Every Lab

1. Open the lab folder: instruction PDF, sample prompts and the mock-data workbook.
2. Save the workbook to OneDrive, turn AutoSave on and open Copilot.
3. Prompt with a named table, columns and expected output; log the prompt.
4. Inspect what Copilot built, then run the Test It checks against the stated values.


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CAPABILITIES (MICROSOFT SUPPORT)

# What Copilot in Excel Can Do

1

## Edit worksheets

Add sheets, set values, apply conditional formatting, validation and styles.

2

## Data and formulas

Generate formula columns, lookups and summary rows - and explain formulas.

3

## Charts and PivotTables

Create and edit charts, PivotTables and dashboards linked to the data.

4

## Insights

Answer questions as summaries, trends, outliers, charts or PivotTables.

5

## Python analysis

Direct answers and advanced analysis mode using Python in the cloud.

6

## Skills and rules

Reusable skills and workbook rules for repeatable processes.


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BEFORE YOU RELY ON A RESULT

# The C34 Verification Gate

1

## Structure

Does each row have one meaning, with unique IDs and consistent categories?

2

## Logic

Do the formula, aggregation, filter and denominator match the KPI definition?

3

## Reconciliation

Does an independent COUNTIFS, SUMIFS or PivotTable total agree?

4

## Visual

Are title, units, scale and categories honest and complete?

5

## Claim

Is each sentence OBSERVED evidence or a HYPOTHESIS to test?

6

## People impact

Is the result aggregated, anonymous and approved by an accountable owner?


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01

TOPIC 01

# Use Copilot to Analyse Data

Copilot in Excel modes and requirements · preparing HR data as tables · prompting for analysis · direct answers, highlight, sort and filter · formula and lookup columns


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# Key Concepts — Use Copilot to Analyse Data

1

## Three modes

Allow editing (Edit) changes the workbook, Plan proposes steps first, Chat only answers without editing.

2

## Requirements

Eligible license, .xlsx saved to OneDrive or SharePoint, AutoSave on, automatic calculation.

3

## Tables first

One header row, one record per row, one meaning per column, a named Excel table.

4

## Specific prompts

Goal, Context, Source and Expectations - and name the exact columns to analyse.

5

## Direct answers

Python-based answers in Chat; outputs inserted from Chat are static, not refreshable.

6

## Verify formulas

Ask Copilot to explain each formula, test edge rows and reconcile to a control total.


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TOPIC 01 · CONCEPT

# What Copilot in Excel Is and How It Works

## WHAT

- Copilot in Excel is an AI assistant, opened from the Copilot button, that plans and carries out workbook tasks in natural language using Excel's own features: formulas, tables, PivotTables, charts, conditional formatting, sorting, filtering and Python. It offers three modes - Allow editing (Edit), Plan and Chat only - and its results stay as ordinary, editable Excel content.

## WHY IT MATTERS

- After a prompt, Copilot builds a step-by-step plan, executes it, then checks whether the result matches your intent; you can watch its reasoning and stop it at any time. Because Edit mode changes a shared workbook directly, choosing the mode deliberately and reviewing Show Changes or version history keeps you in control. Microsoft states it is an assistant, not an autopilot.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — What Copilot in Excel Is and How It Works

1

Open the workbook, select the Copilot button and choose Chat only, Plan or Allow editing.

2

Describe the task; Copilot plans, uses Excel tools, and reports what it did in the pane.

3

Review the cells, objects and changes; keep, refine with a follow-up prompt, or Undo.


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# In Practice — What Copilot in Excel Is and How It Works

## WORKED EXAMPLE

- A BrightPath HR analyst opens the candidate pipeline workbook in Excel for the web.
- Chat only: 'Describe what each row in this table represents' - nothing in the sheet changes.
- Plan, then Allow editing: Copilot proposes and then adds a Days_Open column, which is reviewed.

## DECISION GUIDE

- **Use when**
  - A task maps to Excel features you could inspect and edit yourself afterwards.
  - You can review each change and restore an earlier version if the result is wrong.
- **Avoid when**
  - A decision about an individual (hiring, pay, exit) would rest on an unchecked AI output.
  - The data is not approved for use with Copilot under your organisation's policy.


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# Quality Lens — What Copilot in Excel Is and How It Works

## Mode

- Chat for questions, Plan to preview, Edit only for an approved change.

## Traceability

- Use Show Changes, the Prompt_Log sheet and version history.

## Ownership

- A named person reviews and approves the workbook before it is shared.


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# Requirements, Licensing, Platforms and Data Privacy

## WHAT

- Copilot appears in Excel for Microsoft 365 on Windows, Mac, the web and mobile when the account has an eligible plan: a commercial Microsoft 365 Copilot licence, a Copilot Chat-eligible business plan, or Microsoft 365 Personal, Family or Premium with AI credits. The workbook must be saved to OneDrive or SharePoint with AutoSave on, and Copilot editing needs Automatic calculation.

## WHY IT MATTERS

- Most 'Copilot is missing' problems are setup problems: an unsaved local file, AutoSave off, a Strict Open XML file, a checked-out SharePoint file or a licence or tenant setting. For HR data, work accounts keep prompts and data inside the Microsoft 365 service boundary and respect existing permissions - but the organisation's data policy still decides what personal data may be used.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Requirements, Licensing, Platforms and Data Privacy

1

Check the licence in your Microsoft 365 account; an in-app label may read M365 Copilot (Premium or Basic).

2

Save as .xlsx in OneDrive or SharePoint, turn AutoSave on and set Formulas > Calculation Options > Automatic.

3

Use pseudonymised IDs instead of names, and share only aggregated HR results.


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# In Practice — Requirements, Licensing, Platforms and Data Privacy

## WORKED EXAMPLE

- A learner opens a CSV from Downloads: the Copilot button is grey.
- They save it to OneDrive/work-c34 as .xlsx, AutoSave switches on, and Copilot becomes available.
- Employee names are not needed for analysis, so the workbook uses Employee_ID only.

## DECISION GUIDE

- **Use when**
  - The file is cloud-saved, AutoSave is on and your plan or tenant enables Copilot.
  - HR data is minimised to the columns the analysis actually needs.
- **Avoid when**
  - Edit mode on iPhone or Android, where Chat is currently the supported mode.
  - Pasting sensitive personal data into prompts when an ID or aggregate would do.


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# Quality Lens — Requirements, Licensing, Platforms and Data Privacy

## Setup

- Cloud .xlsx, AutoSave On, Automatic calculation, eligible licence.

## Privacy

- Minimise personal data; pseudonymise; aggregate before sharing.

## Model awareness

- Edit tasks may use Claude or GPT models; outputs still need review.


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# Preparing HR and Recruitment Data for Copilot

## WHAT

- Copilot works with tables, table-like ranges or several such ranges; it does not analyse unstructured data. An analysis-ready HR table has one header row, one record per row (one application, one job order, one employee), one meaning per column and consistent types. Formatting the range as a named Excel table makes its boundary explicit to people, formulas and Copilot.

## WHY IT MATTERS

- Merged headings, blank rows, subtotal rows, duplicated IDs and inconsistent labels such as 'LinkedIn', 'Linked In' and 'LinkedIn ' split categories and corrupt counts. If you cannot say what one row means, no prompt can fix the logic. Recruiting data also changes daily, so a table that expands automatically keeps formulas, PivotTables and charts in step.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Preparing HR and Recruitment Data for Copilot

1

Write the grain: 'one row = one candidate application identified by Application_ID'.

2

Press Ctrl+T (Home > Format as Table), confirm headers, name it e.g. tblApplications.

3

Record row count and unique-ID count; list duplicates and label variants before analysing.


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# In Practice — Preparing HR and Recruitment Data for Copilot

## WORKED EXAMPLE

- BrightPath's pipeline export has 242 rows but only 240 unique Application_IDs.
- Five Source values use non-standard spellings that would split the LinkedIn and JobStreet totals.
- The analyst records these issues and asks Copilot to flag - not delete - them.

## DECISION GUIDE

- **Use when**
  - The data can be expressed as a rectangular list with a stable row meaning.
  - Each column has one data type and a documented list of valid values.
- **Avoid when**
  - A sheet mixes report titles, notes, subtotals and several tables in one range.
  - Fixing an issue would require guessing a missing business fact.


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# Quality Lens — Preparing HR and Recruitment Data for Copilot

## Grain

- One sentence that says exactly what a row represents.

## Schema

- Unique headers, consistent types and canonical category values.

## Controls

- Row count, unique-ID count and key totals recorded before any change.


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# Prompting Copilot for Data Analysis

## WHAT

- Microsoft's prompt ingredients are Goal (what you want), Context (why and for whom), Source (which data) and Expectations (how to respond). This course extends them to G-C-D-C-O-R: Goal, Context, Data, Constraints, Output and Review - so every prompt names the table and columns, what must not change, the output format and how you will check it.

## WHY IT MATTERS

- Microsoft's own guidance is to be specific: naming the columns to analyse produces more accurate results, and 'Compare Q1 and Q2 hires' beats 'Tell me about hiring'. 'Summarise column C' names a task but not the goal. Follow-up prompts keep context, so refine one dimension at a time, and type 'new topic' when you switch tasks.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Prompting Copilot for Data Analysis

1

State one goal and the HR decision it supports.

2

Name the table, columns, filters, period and units; say what must stay unchanged.

3

Ask for a specific output (number, table, column, chart) and for the logic to check it.


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# In Practice — Prompting Copilot for Data Analysis

## WORKED EXAMPLE

- Weak: 'Analyse my recruitment data.'
- Strong: 'Using tblApplications, count applications and hires by Source for Jan-Jun 2026.'
- 'Show the result as a table sorted by hires, and explain how Hired was identified.'

## DECISION GUIDE

- **Use when**
  - You can name the data boundary and the expected output.
  - The answer has a checkable acceptance test (a count, a total, a rule).
- **Avoid when**
  - The request is 'make this better' with no audience, rule or decision.
  - Copilot would have to invent business thresholds or facts not in the data.


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# Quality Lens — Prompting Copilot for Data Analysis

## Specific

- Name the table, columns, period, units and output.

## Bounded

- State exclusions, privacy limits and what must not change.

## Reviewable

- Ask for logic, assumptions and a success check.


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# Direct Answers, Highlighting, Sorting and Filtering

## WHAT

- In Chat, Copilot answers data questions directly using Python-based analysis: it interprets the question, finds the relevant data, computes the answer and can show commented code. Separately, Copilot can apply conditional formatting, sorting and filtering - for example 'Highlight the lowest values in units sold' or 'Sort engagement rate from lowest to highest' - describing each change before you apply it.

## WHY IT MATTERS

- Direct answers do not modify the workbook, and tables or images inserted from them are static - they will not refresh. Highlights, sorts and filters change only what you see, not the data, yet a filter can hide the denominator. Writing the exact rule (field, operator, value) makes the view reproducible and lets a colleague check it.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Direct Answers, Highlighting, Sorting and Filtering

1

Ask a specific question, expand the code or ask Copilot to summarise its logic.

2

Turn the HR question into explicit rules: 'Stage = Hired' or 'Interview_Score >= 4'.

3

Apply one change at a time, then count visible rows and clear filters to restore the view.


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# In Practice — Direct Answers, Highlighting, Sorting and Filtering

## WORKED EXAMPLE

- 'How many applications came from each Source?' returns a direct answer with code.
- 'Highlight rows where Stage is Hired in light green' adds a conditional format rule.
- 'Filter to Technology applications at the Interview stage' shows 9 of 240 records.

## DECISION GUIDE

- **Use when**

– A quick, checkable answer or a reversible review view is needed.

– The rule can be written with named fields, operators and values.

- **Avoid when**

– Colour alone would be the only record of a status.

– A static inserted result will be mistaken for a live figure.


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# Quality Lens — Direct Answers, Highlighting, Sorting and Filtering

## Rule

- Write the condition before applying colour or a filter.

## Denominator

- Record total rows and visible rows together.

## Static vs live

- Label inserted answers 'static as at <date>'.


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# Formula Columns, Lookups and Formula Explanations

## WHAT

- Copilot can add calculated columns or summary rows (totals, durations, margins), suggest a formula for one cell, build lookups such as XLOOKUP to pull matching data from another sheet, and explain any existing formula. In a table it uses structured references such as [@Filled] and [@Fee_Rate] that fill the whole column and adjust as rows are added.

## WHY IT MATTERS

- A formula can be syntactically valid and still wrong: Days_Open may ignore open requisitions, a fee may use the monthly instead of the annual salary, or blanks may silently become zero. Asking Copilot to 'Explain the formula in the selected cell', testing a few rows by hand and reconciling totals turns a generated formula from a guess into evidence.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Formula Columns, Lookups and Formula Explanations

1

Write the business rule in words, including blanks, units and the report date.

2

Ask for one column at a time and for an explanation of the formula.

3

Hand-check two rows (one normal, one edge case) and compare a column total with an independent formula.


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# In Practice — Formula Columns, Lookups and Formula Explanations

## WORKED EXAMPLE

- 'Add a column Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate.'
- 'Create a column that looks up Recruiter_Name from tblRecruiters using Recruiter_ID.'
- JOB-2002: 4 x S$4,900 x 12 x 25% = S$58,800 - matches the generated value.

## DECISION GUIDE

- **Use when**
  - The rule is deterministic and expressible with native Excel functions.
  - A hand-worked sample or control total can verify the result.
- **Avoid when**
  - A subjective judgement is dressed up as a precise calculation.
  - The formula hides missing data or an unapproved threshold.


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# Quality Lens — Formula Columns, Lookups and Formula Explanations

## Semantics

- Check numerator, denominator, units and the date used.

## Edge cases

- Test blanks, zero, open items and one large value.

## Reconcile

- Tie the column total to an independent control.


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WHAT YOU'LL DO

# Hands-On Labs — Use Copilot to Analyse Data

## Lab 1

- Explore and Question Recruitment Pipeline Data

## Lab 2

- Generate and Verify Formula Columns and Lookups for Job Requisitions


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TOPIC 01 · HANDS-ON

# Explore and Question Recruitment Pipeline Data

## LAB 1

You are a talent analyst at BrightPath Talent Solutions, a fictional Singapore recruitment and staffing firm. You will put the H1 2026 candidate-pipeline export into a named table, ask Copilot (Chat only) to describe it, use Copilot to find duplicate IDs and inconsistent Source labels, then answer pipeline questions and build a highlighted, sorted and filtered review view. Every number is checked with a formula.

### You'll build

**BrightPath_Candidate_Pipeline.xlsx in OneDrive with table tblApplications, a Source_Clean column, duplicate IDs highlighted, Hired rows highlighted, a verified source-by-hires answer and a filled Prompt_Log**

| 1 Save to OneDrive | 2 Create the table | 3 Describe the data | 4 Check quality | 5 Ask, highlight, filter |
|---|---|---|---|---|

**Tools:** Excel for Microsoft 365 or Excel for the web · Copilot (Chat only, Allow editing) · BrightPath_Candidate_Pipeline.xlsx


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# Sample Prompts — Lab 1

1. Describe: Describe tblApplications: what one row represents, each column's data type, the row count and the Applied_Date range. Do not change the workbook.

2. Data quality: List any Application_ID that appears more than once in tblApplications and any Source value that is not in the canonical list on About_This_Data.

3. Clean helper column: Add a column Source_Clean that trims spaces from Source and replaces 'Linked In' with 'LinkedIn'. Do not change the original Source column.

4. Question: Using tblApplications, count applications and hires by Source_Clean and show the hire rate. Sort by hires, highest first.

5. Question: Which Client_Sector received the most applications, and how many? Show the count for every sector.


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LAB 1 · VERIFY

# Explore and Question Recruitment Pipeline Data

## ✅ Test it

tblApplications exists with 242 data rows and =COUNTA(UNIQUE(tblApplications[Application_ID])) returns 240. The repeated IDs are APP-0018 and APP-0020 (both Rejected) and are orange. Copilot found 3 'Linked In' and 2 'JobStreet ' (trailing space) values; =COUNTIF(tblApplications[Source_Clean],"LinkedIn") returns 69. Hires total 22 and LinkedIn has the most hires (8). The Technology + Interview filter shows 9 rows. Technology received the most applications (49 unique). After clearing the filter all 242 rows are visible, and Prompt_Log has at least 5 entries.


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TOPIC 01 · HANDS-ON

# Generate and Verify Formula Columns and Lookups for Job Requisitions

## LAB 2

BrightPath's permanent-recruitment desk tracks 60 client job orders. You will ask Copilot to add an XLOOKUP for the recruiter's name and four calculated columns written from business rules, ask it to explain each formula, hand-check one requisition and reconcile column totals to independent formulas on the Control sheet. You will then ask Copilot which recruiter generated the most estimated fees and verify the answer.

### You'll build

**BrightPath_Job_Requisitions.xlsx with Recruiter_Name, Days_Open, Fill_Rate, Est_Fee and Ageing_Flag columns in tblRequisitions, overdue requisitions highlighted, and a Control sheet that reconciles the totals**

| 1 Lookup | 2 Days open | 3 Fill rate & fee | 4 Ageing flag | 5 Reconcile |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Allow editing, Chat only) · XLOOKUP · structured references · BrightPath_Job_Requisitions.xlsx


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# Sample Prompts — Lab 2

1. Lookup: Add a column Recruiter_Name that looks up Recruiter_Name from tblRecruiters using Recruiter_ID with XLOOKUP. Return "Not found" if missing.

2. Duration: Add a column Days_Open: if Close_Date is blank use Control!B1 minus Open_Date, otherwise Close_Date minus Open_Date.

3. Ratio: Add a column Fill_Rate = Filled / Openings formatted as a percentage.

4. Business rule: Add a column Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, formatted as S$ with no decimals.

5. Flag: Add a column Ageing_Flag that shows "Overdue" when Status is Open and Days_Open > 45, otherwise blank.


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LAB 2 · VERIFY

# Generate and Verify Formula Columns and Lookups for Job Requisitions

## ✅ Test it

Control!B3 and B4 both show S$1,030,512. B5 shows 34.6 days (29 Filled requisitions). B6 and B7 both show 13. JOB-2002 shows Est_Fee S$58,800 and Days_Open 36. Every Recruiter_Name resolves (no "Not found"). The top recruiter by estimated fees is Aisha Rahman with S$221,760.


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TOPIC RECAP

# Recap - Use Copilot to Analyse Data

1. LO1 - Analyse data with Copilot: Confirm license, OneDrive/AutoSave and mode; format data as a named table; ask specific questions that name columns; highlight, sort and filter with explicit rules; generate formula and lookup columns and reconcile them to independent checks.


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02

TOPIC 02

# Use Copilot to Plot Charts

Choosing the right chart for an HR question · charts and PivotCharts from a prompt · refining charts with follow-up prompts · advanced visuals with Python in Excel


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# Key Concepts — Use Copilot to Plot Charts

1

## Question first

Comparison, trend, composition, distribution or relationship decides the chart type.

2

## Describe the chart

Name x-axis, y-axis, measure, aggregation, filters and chart type in the prompt.

3

## Pivot behind the chart

Summarise with a PivotTable first, so the chart has a checkable total.

4

## Refine by follow-up

Titles, labels, sort order, colours and axis scales are prompt-able edits.

5

## Python visuals

=PY cells run in the Microsoft cloud with Anaconda libraries for box plots and heatmaps.

6

## Honest charts

Units, full categories, zero-based bars and a source note on every chart.


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# Choosing the Right Chart for an HR Question

## WHAT

- Every chart answers one kind of question. Comparison across categories uses a column or bar chart; change over time uses a line chart; a recruitment funnel uses a funnel or ordered bar chart; distribution of salaries or rates uses a histogram or box plot; the relationship between two measures uses a scatter plot; and share of a whole uses a stacked bar or, sparingly, a pie.

## WHY IT MATTERS

- Copilot will draw whatever you ask for, quickly and polished - but a pie chart of 18 months, or an average of averages, can mislead an HR leadership team. Deciding the question, the grain and the measure before prompting is the analyst's job; Copilot then removes the clicking.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Choosing the Right Chart for an HR Question

1

Write the question: 'Which client sector produced the most hires in H1 2026?'

2

Classify it (comparison) and choose the form (sorted bar chart).

3

Name the measure and aggregation (Sum of Hires) and the filter (2026 only).


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# In Practice — Choosing the Right Chart for an HR Question

## WORKED EXAMPLE

- Trend: monthly hires Jan 2025 - Jun 2026 as a line chart.
- Funnel: Applications > Screened > Interviews > Offers > Hires for H1 2026.
- Efficiency: hire rate and cost per hire by sourcing channel as a bar or combo chart.

## DECISION GUIDE

- **Use when**
  - The audience needs one clear message from the visual.
  - The categories and time order are complete and consistent.
- **Avoid when**
  - A pie with many slices or a 3-D effect would distort comparison.
  - Rates are averaged row by row instead of recomputed from totals.


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# Quality Lens — Choosing the Right Chart for an HR Question

## Fit

- Chart type matches comparison, trend, funnel, distribution or relationship.

## Message

- Title states the finding, e.g. 'Technology led H1 hires'.

## Integrity

- Zero-based bars, units shown, no hidden categories.


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# Creating Charts and PivotTables from a Prompt

## WHAT

- Describe the chart you want and Copilot generates it, linked to the source data so it stays editable. Microsoft's examples include 'Create a line chart showing change over time. Show data labels and gridlines.' and 'Create a PivotTable with the sum of sales, with month in rows and category in columns.' The more detail you include, the more useful the result.

## WHY IT MATTERS

- A PivotTable summarises large data sets without manual steps and gives the chart a visible, checkable summary. If the first result does not match your intent, add detail - the field, aggregation, filter or chart type - and ask again rather than starting from scratch.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Creating Charts and PivotTables from a Prompt

1

Prompt for the PivotTable: rows, columns, values, aggregation and filters.

2

Prompt for the chart from that PivotTable and say where to place it (the Charts sheet).

3

Check the Pivot grand total against a SUM or SUMIFS of the source table.


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# In Practice — Creating Charts and PivotTables from a Prompt

## WORKED EXAMPLE

- 'Create a PivotTable from tblMonthly with Month_Label in rows, Sector in columns and Sum of Hires.'
- 'Create a line chart of total hires by month from that PivotTable, with data labels.'
- Check: Pivot grand total for 2026 = 361 hires = SUMIFS(tblMonthly[Hires], ...).

## DECISION GUIDE

- **Use when**
  - The data is in a table and the measure has a defined aggregation.
  - You need an editable chart that updates when the PivotTable refreshes.
- **Avoid when**
  - The question, denominator or aggregation has not been agreed yet.
  - A static image from Chat would be pasted into a recurring report.


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# Quality Lens — Creating Charts and PivotTables from a Prompt

## Grain

- Know what one Pivot cell represents.

## Tie-out

- Pivot grand total equals the source-table control.

## Editable

- Chart is linked to data, not a pasted picture.


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# Refining and Formatting Charts with Follow-up Prompts

## WHAT

- Follow-up prompts keep the conversation's context, so you can refine a chart step by step: change the type, sort the bars, add a secondary axis, rename the title to state the finding, format axis numbers as S$ or %, apply consistent colours and add alt text. Each refinement is still an ordinary Excel chart setting you can inspect in Format Chart.

## WHY IT MATTERS

- First drafts rarely communicate. A combo chart that puts cost per hire (S$) and hire rate (%) on one axis is unreadable; an unsorted bar chart hides the ranking; a generic title such as 'Sum of Hires' tells leadership nothing. Small, specific follow-ups turn a default chart into a decision visual.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Refining and Formatting Charts with Follow-up Prompts

1

Ask for one refinement per prompt: type, sort, axis, labels, title or colours.

2

Use a secondary axis only when measures have different units, and label both axes.

3

Add a source note and alt text, then check the chart against the PivotTable values.


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# In Practice — Refining and Formatting Charts with Follow-up Prompts

## WORKED EXAMPLE

- 'Sort the bars from highest to lowest hire rate.'
- 'Change the title to: Referrals convert best - hire rate by source, Jan-Jun 2026.'
- 'Put Cost per Hire on a secondary axis as a line and format it as S$.'

## DECISION GUIDE

- **Use when**
  - The chart is right in substance but not yet clear to its audience.
  - You want consistent formatting across several charts or a dashboard.
- **Avoid when**
  - A refinement would change the meaning, e.g. truncating a bar axis.
  - Many edits are bundled into one vague prompt such as 'make it pretty'.


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# Quality Lens — Refining and Formatting Charts with Follow-up Prompts

## Readable

- Finding-style title, labelled axes, units and legend.

## Accessible

- Colour-blind-safe palette and alt text.

## Accurate

- Values still match the PivotTable after formatting.


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# Advanced Visuals with Python in Excel

## WHAT

- Python in Excel runs Python code in worksheet cells: type =PY or use Formulas > Insert Python. The code runs in the Microsoft cloud on a standard Anaconda distribution, so no local install is needed, and results return to the grid. Copilot can write the Python for you - including box plots, histograms, heatmaps and regression lines that native charts cannot easily produce.

## WHY IT MATTERS

- HR questions are often about spread, not averages: pay-rate ranges by job level, outliers, or whether experience relates to pay. Python visuals show distributions honestly. But Python in Excel needs internet access, is available on Windows, Mac and the web (not mobile), and its code and data choices must be reviewed like any formula.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Advanced Visuals with Python in Excel

1

Ask Copilot for the visual and to use Python, naming the table and columns.

2

Open the generated =PY cell and read the code: data range, grouping, filters.

3

Cross-check one plotted statistic (e.g. a median) with an Excel formula.


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# In Practice — Advanced Visuals with Python in Excel

## WORKED EXAMPLE

- 'Using Python, create a box plot of Bill_Rate_Hourly by Job_Level in tblPlacements.'
- The Lead median bill rate in the plot is checked with MEDIAN(IF(...)) in Excel.
- 'Create a heatmap of average gross margin % by Sector and Job_Level.'

## DECISION GUIDE

- **Use when**
  - The question is about distribution, correlation or many groups at once.
  - Your licence and platform support Python in Excel and a reviewer can read the code.
- **Avoid when**
  - The workbook will be used offline or on a phone or tablet.
  - A simple native column or line chart would answer the question.


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# Quality Lens — Advanced Visuals with Python in Excel

## Inputs

- Code references the right table, columns and filters.

## Method

- Statistic (median, IQR, correlation) suits the question.

## Reproducible

- One plotted value re-computed with a native formula.


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WHAT YOU'LL DO

# Hands-On Labs — Use Copilot to Plot Charts

## Lab 3

- Create and Refine Recruitment Charts with Copilot

## Lab 4

- Visualise Pay and Placement Distributions with Python in Excel


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TOPIC 02 · HANDS-ON

# Create and Refine Recruitment Charts with Copilot

## LAB 3

BrightPath's leadership wants to see how hiring is moving and which sectors and sourcing channels perform best. You will ask Copilot to build PivotTables and charts from 18 months of funnel metrics and a sourcing summary, choose each chart from the question it answers, refine titles, labels, sort order and axes with follow-up prompts, and reconcile every chart to an independent SUMIFS check.

### You'll build

**BrightPath_Recruitment_Metrics.xlsx with PivotTables and four refined charts on the Charts sheet: a monthly hires line, a sorted 2026 hires-by-sector bar, a 2026 recruitment funnel and a hire-rate vs cost-per-hire combo chart**

| 1 Trend line | 2 Sector bars | 3 Funnel | 4 Source combo | 5 Refine & check |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Allow editing) · PivotTables · line, bar, funnel and combo charts · BrightPath_Recruitment_Metrics.xlsx


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# Sample Prompts — Lab 3

1. PivotTable: Create a PivotTable from tblMonthly with Month_Label in rows, Sector in columns and Sum of Hires as values.

2. Line chart: Create a line chart showing total hires by month. Show data labels and gridlines.

3. Bar chart: Create a bar chart of 2026 hires by Sector, sorted from highest to lowest.

4. Funnel: Create a funnel chart of 2026 Applications, Screened, Interviews, Offers and Hires with data labels.

5. Combo chart: Create a combo chart of Hire_Rate as columns and Cost_per_Hire as a line on a secondary axis.


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LAB 3 · VERIFY

# Create and Refine Recruitment Charts with Copilot

## ✅ Test it

Pivot_Trend grand total = 1,065 hires (= SUM(tblMonthly[Hires])); the peak month is 2025-11 with 68 hires. The 2026 sector bar chart shows Technology first with 84 hires and its Pivot total is 361. The funnel shows 6,616 > 3,229 > 1,385 > 495 > 361. In tblSources, Referral has the highest hire rate (7.4%) and MyCareersFuture the lowest cost per hire (S$186). All four charts sit on the Charts sheet with finding-style titles, labelled axes and alt text.


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TOPIC 02 · HANDS-ON

# Visualise Pay and Placement Distributions with Python in Excel

## LAB 4

BrightPath's flexible-staffing desk places contract associates with clients. You will ask Copilot to add a gross margin column, then create Python in Excel visuals - a box plot of bill rates by job level, a scatter of experience against pay with a trend line, and a sector-by-level margin heatmap. You will read the =PY code, re-compute a median and a correlation with Excel formulas, and investigate the highest pay rate.

### You'll build

**BrightPath_Contract_Placements.xlsx with a Gross_Margin_Pct column and, on Python_Charts, a box plot, a scatter with trend line, a margin heatmap and an outlier table - each with a native-formula check beside it**

| 1 Margin column | 2 Box plot | 3 Scatter | 4 Heatmap | 5 Outlier check |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Allow editing, Chat only) · Python in Excel (=PY) · BrightPath_Contract_Placements.xlsx


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# Sample Prompts — Lab 4

1. Calculated column: Add Gross_Margin_Pct = (Bill_Rate_Hourly - Pay_Rate_Hourly) / Bill_Rate_Hourly as a percentage.

2. Box plot: Using Python, create a box plot of Bill_Rate_Hourly by Job_Level ordered Junior, Mid, Senior, Lead.

3. Histogram: Using Python, create a histogram of Pay_Rate_Hourly with 15 bins and mark the median.

4. Scatter: Using Python, plot Years_Experience against Pay_Rate_Hourly with a linear trend line and return the correlation.

5. Heatmap: Using Python, create a heatmap of average Gross_Margin_Pct by Sector and Job_Level with values shown.


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LAB 4 · VERIFY

# Visualise Pay and Placement Distributions with Python in Excel

## ✅ Test it

Average Gross_Margin_Pct = 22.6%. The box plot shows four levels in order and the Lead median bill rate check returns S$65.74 (Junior S$22.72, Mid S$29.53, Senior S$41.70). CORREL returns about 0.49 - a moderate positive relationship. The heatmap shows Technology with the lowest average margin (~19.1%). The IQR rule flags 8 pay-rate outliers above about S$68.44; the highest is PL-5060 at S$173.47, and your Prompt_Log records a decision about it.


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TOPIC RECAP

# Recap - Use Copilot to Plot Charts

1. LO2 - Plot charts with Copilot: Match the HR question to comparison, trend, composition, distribution or relationship; prompt for the exact fields, aggregation and chart type; refine titles, labels and axes; and tie every chart back to a PivotTable or source total.


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03

TOPIC 03

# Use Copilot to Create Dashboards

HR and recruitment KPI design · building a dashboard with Plan and Edit modes · slicers, timelines and refresh · sharing people dashboards responsibly


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# Key Concepts — Use Copilot to Create Dashboards

1

## KPI definitions

Time to hire, offer acceptance, cost per hire, fill rate, headcount and attrition - defined first.

2

## Plan before build

Ask Copilot to plan the dashboard, review it, then let it build in Edit mode.

3

## Dashboard anatomy

KPI cards on top, 3-4 charts in the middle, slicers at the side, source note below.

4

## Interactivity

Slicers and timelines filter every PivotTable and chart connected to them.

5

## Refreshable by design

Tables + PivotTables + formulas refresh; pasted images and static answers do not.

6

## Responsible sharing

Aggregate, suppress small groups, control permissions and track changes.


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TOPIC 03 · CONCEPT

# Designing HR and Recruitment KPIs

## WHAT

- A KPI is a defined measure tied to a decision, with a formula, a period and a target. Common recruitment KPIs are applications, hires, time to hire, offer acceptance rate and cost per hire; workforce KPIs are headcount, new hires, attrition rate and average tenure. Microsoft's HR adoption scenarios cite cost per hire, onboarding time, retention and eNPS as measures Copilot can help improve.

## WHY IT MATTERS

- Two people can compute 'offer acceptance rate' differently - including or excluding pending offers - and both will look right. Writing each definition before building lets you check Copilot's formula against it and keeps the dashboard consistent month after month.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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TOPIC 03 · PROCESS

# How It Works — Designing HR and Recruitment KPIs

1

List the 4-6 decisions the dashboard supports and one KPI for each.

2

Write the formula, period and exclusions for each KPI in a definitions note.

3

Pick the dimensions users will filter by: month, sector, source, recruiter, department.


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# In Practice — Designing HR and Recruitment KPIs

## WORKED EXAMPLE

- Offer acceptance rate = accepted / (accepted + declined); pending offers excluded.
- Cost per hire = total Sourcing_Cost / hires in the period.
- Attrition rate = exits in 12 months / average of opening and closing headcount.

## DECISION GUIDE

- **Use when**
  - The same measures are reviewed regularly by the same audience.
  - Each KPI has an owner who can act on a change.
- **Avoid when**
  - A metric has no agreed definition or no one acts on it.
  - Too many KPIs crowd the page and bury the important ones.


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# Quality Lens — Designing HR and Recruitment KPIs

## Defined

- Formula, period and exclusions written down.

## Actionable

- Every KPI maps to a decision and an owner.

## Focused

- Four to six headline KPIs, not twenty.


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# Building a Dashboard with Copilot Plan and Edit Modes

## WHAT

- Copilot can create and edit charts, PivotTables and shapes and build dashboards, planning tools and templates from scratch, with editable links back to the source data. In Plan mode it outlines the sheets, ranges, formulas and objects it intends to create and can ask clarifying questions; in Allow editing mode it carries out that multi-step plan and checks the result.

## WHY IT MATTERS

- A dashboard is a multi-step build - KPI formulas, several PivotTables, charts, slicers and layout - which is exactly where an agentic plan-execute-verify loop saves time. Reviewing the plan first catches a wrong KPI definition before it is copied into five objects.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Building a Dashboard with Copilot Plan and Edit Modes

1

Plan: give the KPI definitions, the dimensions, the target sheet and the layout.

2

Approve or correct the plan, then switch to Allow editing and let Copilot build it.

3

Inspect each KPI card formula and reconcile it to its PivotTable or source total.


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# In Practice — Building a Dashboard with Copilot Plan and Edit Modes

## WORKED EXAMPLE

- 'Plan a recruitment dashboard on the Dashboard sheet using tblApps: KPI cards for applications, hires, offer acceptance rate, average time to hire and cost per hire; charts for hires by month vs target, hires by sector and applications by source; slicers for Sector and Recruiter.'
- The plan is checked against the KPI definitions in About_This_Data before approval.

## DECISION GUIDE

- **Use when**
  - The data is a clean table and the KPI definitions are written down.
  - You can review the plan and each created object afterwards.
- **Avoid when**
  - The workbook is shared live with others and you have not saved a version.
  - You would accept the KPI values without checking a single one.


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# Quality Lens — Building a Dashboard with Copilot Plan and Edit Modes

## Plan review

- KPI definitions in the plan match the written ones.

## Linked

- Cards use formulas or GETPIVOTDATA, not typed numbers.

## Verified

- Each KPI ties to an independent check.


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TOPIC 03 · CONCEPT

# Interactivity, Refresh and Maintenance

## WHAT

- Slicers and timelines are on-sheet filter buttons that can control several PivotTables and charts at once. Because the source is an Excel table, new rows are included automatically; Data > Refresh All updates the PivotTables and charts. Copilot can also store structure, naming and formula conventions as workbook rules, and reusable skills, so the dashboard is rebuilt the same way each month.

## WHY IT MATTERS

- A dashboard that only works for one month's data is a report, not a dashboard. Testing refresh with new rows proves the formulas and PivotTables are linked correctly - and exposes any static images or typed numbers that Copilot or a user inserted.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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TOPIC 03 · PROCESS

# How It Works — Interactivity, Refresh and Maintenance

1

Ask Copilot to connect slicers to all PivotTables on the dashboard.

2

Append test rows to the source table, then Data > Refresh All.

3

Confirm every KPI and chart changed by the expected amount; fix any that did not.


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# In Practice — Interactivity, Refresh and Maintenance

## WORKED EXAMPLE

- Eight July 2026 new joiners are appended to tblEmployees.
- After Refresh All, active headcount rises by exactly 8 and every chart updates.
- A pasted chart image did not update, so it is replaced with a linked PivotChart.

## DECISION GUIDE

- **Use when**
  - The dashboard will be reused with new data each period.
  - Users need to explore by department, sector, recruiter or month.
- **Avoid when**
  - Slicers are left unconnected to some charts, giving inconsistent views.
  - Numbers are typed into KPI cards instead of calculated.


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# Quality Lens — Interactivity, Refresh and Maintenance

## Connected

- Each slicer filters every related object.

## Refresh-tested

- Added rows change KPIs by the expected amount.

## Documented

- Definitions, source and refresh date shown on the sheet.


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# Sharing People Dashboards Responsibly

## WHAT

- People data - salaries, performance, engagement and exits - is sensitive. A responsible HR dashboard shows aggregated results, suppresses groups too small to protect anonymity (for example fewer than 5 people), limits access through OneDrive or SharePoint permissions, and keeps a record of who changed what. Copilot's Show Changes pane attributes edits made by Copilot alongside collaborators.

## WHY IT MATTERS

- Copilot does not give anyone access to data they could not already open - but anyone who can open the workbook sees everything in it, including hidden sheets. Saved changes are visible to co-authors. Responsible sharing protects employees and keeps the dashboard trusted.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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TOPIC 03 · PROCESS

# How It Works — Sharing People Dashboards Responsibly

1

Remove or pseudonymise identifying columns and keep raw data in a restricted file if needed.

2

Check that every slicer combination still shows groups of at least 5 people.

3

Share with specific people, review Show Changes and keep version history.


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# In Practice — Sharing People Dashboards Responsibly

## WORKED EXAMPLE

- Salary detail stays in the source file; the leadership copy shows only averages by level.
- Filtering IT + Director would show one person, so that slicer combination is removed.
- Copilot can then help draft the summary email or slides from the verified dashboard.

## DECISION GUIDE

- **Use when**
  - The dashboard leaves the HR team or is used in a leadership meeting.
  - The data includes pay, performance, health or exit details.
- **Avoid when**
  - Individual-level records are visible to a broad audience.
  - A small group's result could identify a person.


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# Quality Lens — Sharing People Dashboards Responsibly

## Minimise

- Only the columns the audience needs.

## Protect

- Small-group suppression and specific-people sharing.

## Audit

- Show Changes and version history retained.


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WHAT YOU'LL DO

# Hands-On Labs — Use Copilot to Create Dashboards

## Lab 5

- Build a Recruitment KPI Dashboard with Copilot

## Lab 6

- Build an Interactive Workforce and Attrition Dashboard


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TOPIC 03 · HANDS-ON

# Build a Recruitment KPI Dashboard with Copilot

## LAB 5

BrightPath's Head of Recruitment wants one page that shows H1 2026 applications, hires, offer acceptance rate, time to hire and cost per hire, by month, sector, source and recruiter. You will write the checks first, ask Copilot to plan the dashboard, review the plan against the KPI definitions, let Copilot build it in Allow editing mode, and then prove each KPI and slicer works.

### You'll build

**BrightPath_Recruitment_Dashboard_Data.xlsx with a Checks sheet and a Dashboard sheet holding five formula-driven KPI cards, charts for hires vs target by month, hires by sector and applications by source, and Sector and Recruiter slicers connected to every PivotTable**

| 1 Define KPIs | 2 Build checks | 3 Plan with Copilot | 4 Build in Edit | 5 Verify & slice |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Plan, Allow editing) · PivotTables, PivotCharts, slicers · BrightPath_Recruitment_Dashboard_Data.xlsx


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# Sample Prompts — Lab 5

1. Plan: Plan a recruitment KPI dashboard on the Dashboard sheet from tblApps with KPI cards for applications, hires, offer acceptance rate, average time to hire and cost per hire. List the formulas before building.

2. KPI card: Create a KPI card that shows Cost per hire = total Sourcing_Cost / hires, formatted as S$ with 2 decimals.

3. Chart vs target: Create a column chart of hires by Hire_Date month with Hire_Target from tblTargets as a line.

4. Slicers: Add slicers for Sector and Recruiter and connect them to all PivotTables on Dash_Data.

5. Fix a KPI: The offer acceptance rate must exclude Pending offers. Update the formula and explain it.


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LAB 5 · VERIFY

# Build a Recruitment KPI Dashboard with Copilot

## ✅ Test it

With no slicer selected the KPI cards match Checks: Applications 600, Hires 41, Offer acceptance 78.8% (41 accepted of 52 decided; 10 pending excluded), Avg time to hire 27.8 days, Cost per hire S$348.20. Hires by month: 2026-02 = 9, 2026-03 = 10, 2026-04 = 9, 2026-05 = 7, 2026-06 = 6 (none in 2026-01). Manufacturing has the most hires (11). With Sector = Technology, Applications = 124, Hires = 5 and Avg time to hire = 42.0 days. Siti Hamid made the most hires (13). No KPI card contains a typed number.


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TOPIC 03 · HANDS-ON

# Build an Interactive Workforce and Attrition Dashboard

## LAB 6

BrightPath's HR Director needs a workforce dashboard for the July leadership meeting. You will ask Copilot to add period helper columns, type independent checks for headcount and attrition, have Copilot plan and build the dashboard with slicers, then append eight July new joiners and refresh to prove the dashboard is live. Finally you will check that no slicer combination exposes a group too small to stay anonymous.

### You'll build

**BrightPath_Workforce_Headcount.xlsx with helper columns in tblEmployees, a Checks area, and an HR_Dashboard sheet showing six KPI cards, headcount and attrition-rate charts by department, hires vs exits by quarter, and Location and Job_Level slicers - refresh-tested with new rows**

| 1 Helper columns | 2 Checks | 3 Build dashboard | 4 Refresh test | 5 Privacy check |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Plan, Allow editing) · PivotTables, slicers, Refresh All · BrightPath_Workforce_Headcount.xlsx


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# Sample Prompts — Lab 6

1. Helper column: Add Active_Start = TRUE if Hire_Date <= 2025-07-01 and Exit_Date is blank or after 2025-07-01.

2. Helper column: Add Tenure_Years from Hire_Date to Exit_Date, or to 2026-06-30 if Exit_Date is blank, rounded to 1 decimal.

3. Plan: Plan an HR workforce dashboard with KPI cards for active headcount, exits, attrition rate, voluntary attrition, average tenure and hires.

4. Chart: Create a bar chart of 12-month attrition rate by Department, sorted highest first, with data labels as percentages.

5. Slicers: Add slicers for Location and Job_Level and connect them to every PivotTable on the dashboard.


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LAB 6 · VERIFY

# Build an Interactive Workforce and Attrition Dashboard

## ✅ Test it

Before the refresh: Active headcount 237, headcount on 2025-07-01 210, exits in period 29 (25 voluntary), attrition rate 13.0% and voluntary attrition 11.2%, hires in period 56, average tenure of active staff about 3.1 years. Recruitment has the highest attrition rate (21.8%, 12 exits) and the largest active headcount (60). After appending the July joiners and Refresh All, Active headcount shows 245 and tblEmployees has 428 rows. The dashboard shows no Employee_ID, salary or individual record.


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TOPIC RECAP

# Recap - Use Copilot to Create Dashboards

1. LO3 - Create dashboards with Copilot: Define each KPI before building; ask Copilot to plan, then build KPI cards, PivotTables, charts and slicers; prove the dashboard refreshes when rows are added; and share only aggregated, permissioned views.


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04

TOPIC 04

# Use Copilot to Draw Insights and Recommendations

Surfacing trends, outliers and drivers · advanced analysis mode with Python · themes and sentiment in feedback · from verified insight to owned recommendation


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# Key Concepts — Use Copilot to Draw Insights and Recommendations

1

## Insight ladder

Observation, evidence, implication, recommendation, owner and KPI.

2

## What's interesting?

A good opening prompt - followed by specific, checkable follow-ups.

3

## Advanced analysis

Python on a new sheet for deeper, editable, refreshable analysis.

4

## Text analysis

Themes and sentiment from comments as a new column, spot-checked by a human.

5

## Observed vs hypothesis

A pattern is not a cause; label explanations as hypotheses to test.

6

## Human decision

People decisions stay with people; Copilot informs, it does not decide.


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# Surfacing Trends, Outliers and Drivers

## WHAT

- Ask Copilot about your data in plain language and it surfaces summaries, trends and outliers as text, charts or PivotTables. Microsoft's examples include 'What's interesting about this data?', 'Are there any outliers or anomalies in this data?' and 'What are the mean and median of revenue by region?'. A driver analysis compares an outcome, such as attrition, across groups or measures.

## WHY IT MATTERS

- Open questions are a good start, but they can return selective or causal-sounding stories. Specific follow-ups with a baseline and denominator - 'attrition rate for engagement below 3 versus 3 and above, with counts' - produce evidence you can check. Small groups (an IT team of 25) can show extreme rates by chance.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Surfacing Trends, Outliers and Drivers

1

Start broad ('What's interesting?'), then ask for specific comparisons with counts.

2

Always ask for the denominator and group size behind each rate.

3

Verify the headline numbers with a PivotTable or COUNTIFS before recording them.


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# In Practice — Surfacing Trends, Outliers and Drivers

## WORKED EXAMPLE

- Overall 12-month attrition in the study group: 56 of 400 = 14.0%.
- Engagement below 3: 29.6% left (n = 81) vs 10.0% (n = 319) at 3 and above.
- IT shows 28.0% - but only 25 people, so it is flagged as a small-sample signal.

## DECISION GUIDE

- **Use when**
  - You need candidate patterns to investigate further.
  - Each claimed pattern can be traced to rows and a calculation.
- **Avoid when**
  - A short period or small group is used to claim a lasting trend.
  - A coincident pattern is presented as the proven cause.


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# Quality Lens — Surfacing Trends, Outliers and Drivers

## Baseline

- Show the comparison group, denominator and n.

## Method

- State how a trend or outlier was defined.

## Language

- OBSERVED for evidence, HYPOTHESIS for explanations.


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# Advanced Analysis Mode with Python

## WHAT

- After a direct answer, choose 'Get deeper results using advanced analysis mode' or ask Copilot to 'Enter advanced analysis mode'. Copilot opens a new sheet and runs Python for deeper, customisable analysis - group comparisons, correlations, clustering or regression - which you can modify and insert as a refreshable Python cell. Select 'Stop advanced analysis' to return.

## WHY IT MATTERS

- Python widens what you can ask - 'Which factors are most associated with leaving?' - but correlation is not causation, and the model's choices (which rows, how blanks are handled, which test) shape the answer. Microsoft advises reviewing the commented code and asking Copilot to summarise it to check its logic.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Advanced Analysis Mode with Python

1

Ask a precise question naming the table, outcome, candidate factors and output.

2

Read the code: filters, grouping, missing values and the statistic used.

3

Re-compute one headline value in Excel and record the method and limitations.


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# In Practice — Advanced Analysis Mode with Python

## WORKED EXAMPLE

- 'Compare employees who left with those who stayed on Overtime_Hours_Month, Engagement_Score, Years_Since_Promotion and Compa_Ratio. Show means and counts.'
- Leavers average 15.9 overtime hours a month vs 10.6 for stayers - checked with AVERAGEIFS.

## DECISION GUIDE

- **Use when**
  - The question needs statistics beyond a PivotTable.
  - A reviewer can read the code and reproduce the headline numbers.
- **Avoid when**
  - The output would be used to predict or decide about a named individual.
  - Nobody can explain the method used.


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# Quality Lens — Advanced Analysis Mode with Python

## Inputs

- Right table, rows, columns and blank handling.

## Method

- Statistic fits the question; association not causation.

## Reproducible

- Headline values re-computed with native formulas.


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# Analysing Survey Comments: Themes and Sentiment

## WHAT

- For text such as survey results, reviews or feedback, Copilot can summarise comments, identify themes and sentiment, and add the results as a new column. In the grid, the COPILOT function - for example =COPILOT("Classify the theme of this comment into one of", D2, F2:F8) - classifies text row by row where your licence and channel support it. Hover over a superscript citation to see which rows Copilot used.

## WHY IT MATTERS

- Hundreds of comments can be themed in minutes, which makes employee voice usable in an engagement review. But AI classification can be wrong, inconsistent between runs, and is not suited to exact calculation - Microsoft recommends native functions such as SUM, AVERAGE and IF for numbers. A fixed theme list and a human spot-check keep it honest.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Analysing Survey Comments: Themes and Sentiment

1

Give Copilot a fixed list of themes and sentiments to choose from, plus 'Unclear'.

2

Add the classification as new columns; keep the original comment unchanged.

3

Spot-check at least 10 rows; if more than 1 in 10 is wrong, refine the prompt and rerun.


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# In Practice — Analysing Survey Comments: Themes and Sentiment

## WORKED EXAMPLE

- Themes: Workload, Career Growth, Manager Support, Pay & Benefits, Recognition, Tools & Systems, Flexibility.
- Workload is the most frequent theme in BrightPath's June pulse survey.
- eNPS itself is computed with COUNTIFS, not with the COPILOT function.

## DECISION GUIDE

- **Use when**
  - Comments are many and the theme list is agreed in advance.
  - A person reviews a sample and owns the final counts.
- **Avoid when**
  - Comments could identify the respondent and would be shown verbatim.
  - The AI label would be used as an exact metric without review.


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# Quality Lens — Analysing Survey Comments: Themes and Sentiment

## Taxonomy

- Fixed theme list plus Unclear.

## Spot-check

- Sample reviewed and error rate recorded.

## Anonymity

- Do not quote comments that reveal a person.


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# From Verified Insight to Owned Recommendation

## WHAT

- A recommendation turns a verified finding into an action: Finding (OBSERVED) - Evidence (cells and numbers) - Implication - Recommendation - Owner - Priority - Success KPI and review date. Copilot can draft each part, an executive summary, and - through Copilot in Word, PowerPoint or Outlook - the memo, slides or email that communicates it.

## WHY IT MATTERS

- AI output can contain mistakes; Microsoft advises reviewing, editing and verifying everything before relying on it, and not relying on it alone for high-stakes decisions. Recommendations about people affect livelihoods: they should target teams, processes and policies, cite evidence, state limitations and be approved by an accountable person.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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TOPIC 04 · PROCESS

# How It Works — From Verified Insight to Owned Recommendation

1

Select 3-5 verified findings and write each as an OBSERVED statement with numbers.

2

Ask Copilot to propose actions, owners and KPIs; edit them to fit your organisation.

3

Draft the summary, check every number against the workbook, then decide Share, Revise or Hold.


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# In Practice — From Verified Insight to Owned Recommendation

## WORKED EXAMPLE

- Finding: Workload balance is the lowest-scoring item (2.71 of 5) and the top comment theme.
- Recommendation: review recruiter desk targets and add peak-season support - owner: Head of Recruitment.
- KPI: raise Q3_Workload_Balance to 3.2 and eNPS above 0 by the December pulse.

## DECISION GUIDE

- **Use when**
  - Findings are verified and an accountable owner exists.
  - The recommendation can be measured and reviewed.
- **Avoid when**
  - An action targets an identifiable individual based on model output.
  - A hypothesis is presented as a proven cause.


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# Quality Lens — From Verified Insight to Owned Recommendation

## Evidence-backed

- Every recommendation cites a verified finding.

## Owned

- Named owner, priority and review date.

## Measurable

- A KPI and target that will show if it worked.


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WHAT YOU'LL DO

# Hands-On Labs — Use Copilot to Draw Insights and Recommendations

## Lab 7

- Surface Attrition Insights with Copilot and Advanced Analysis

## Lab 8

- Turn Engagement Survey Feedback into Recommendations


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TOPIC 04 · HANDS-ON

# Surface Attrition Insights with Copilot and Advanced Analysis

## LAB 7

BrightPath lost 14% of a 400-person study group in 12 months. You will start with an open 'What's interesting?' prompt, set the baseline attrition rate, ask Copilot for specific group comparisons with counts, enter advanced analysis mode to compare leavers and stayers with Python, re-compute the headline numbers with COUNTIFS and AVERAGEIFS, and write four labelled insights - including why a high rate in a small team is only a signal.

### You'll build

**BrightPath_Attrition_Analysis.xlsx with verified attrition rates by department and factor, an advanced-analysis sheet with reviewed Python output, and an Insight_Log of four OBSERVED or HYPOTHESIS insights with evidence and limitations**

| 1 Ask broadly | 2 Set the baseline | 3 Compare groups | 4 Advanced analysis | 5 Log insights |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Chat only, advanced analysis mode) · Python in Excel · BrightPath_Attrition_Analysis.xlsx


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# Sample Prompts — Lab 7

1. Open question: What's interesting about this data in relation to Left_Company? Give five patterns with numbers and group sizes.

2. Baseline: What is the overall attrition rate in tblAttrition? Show the numerator and denominator.

3. Comparison: Show attrition rate and headcount for Engagement_Score below 3 vs 3 and above.

4. By group: Show attrition rate and number of employees by Department, sorted highest first.

5. Advanced analysis: Enter advanced analysis mode. Compare leavers and stayers on overtime, engagement, promotion, compa-ratio and training.


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LAB 7 · VERIFY

# Surface Attrition Insights with Copilot and Advanced Analysis

## ✅ Test it

Overall attrition = 56 of 400 = 14.0%. Engagement below 3: 29.6% (n = 81) vs 10.0% (n = 319). Overtime above 20 hours: 34.0% (n = 47) vs 11.3% (n = 353). Promotion 3+ years ago: 21.3% vs 13.0%. On-site staff: 19.3%. IT shows the highest department rate (28.0%) but has only 25 people, so it is recorded as a small-sample signal. Leavers average 15.9 overtime hours and engagement 3.10, vs 10.6 and 3.54 for stayers. Overtime and engagement correlate at about -0.25, so the two factors overlap. Insight_Log has four rows, each labelled and with a limitation.


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TOPIC 04 · HANDS-ON

# Turn Engagement Survey Feedback into Recommendations

## LAB 8

BrightPath ran an anonymous pulse survey of 160 staff. You will have Copilot calculate eNPS and question averages and verify them with COUNTIFS and AVERAGE, compare departments while respecting small groups, add Theme and Sentiment columns from a fixed list and spot-check them, then use the Recommendations template and Copilot to turn verified findings into owned actions and an executive summary you approve.

### You'll build

**BrightPath_Engagement_Survey.xlsx with verified eNPS and question averages, Theme and Sentiment columns with a recorded spot-check, a Recommendations sheet with three owned, measurable actions, and an Executive_Summary**

| 1 Score & eNPS | 2 By department | 3 Theme comments | 4 Recommend | 5 Summarise |
|---|---|---|---|---|

**Tools:** Excel · Copilot (Chat only, Allow editing) · COPILOT function (where available) · BrightPath_Engagement_Survey.xlsx


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# Sample Prompts — Lab 8

1. Score: Calculate eNPS from eNPS_Score in tblSurvey: promoters 9-10, passives 7-8, detractors 0-6. Show the counts.

2. Averages: Give the average of each question Q1 to Q6, lowest to highest, with the number of responses.

3. Groups: Show eNPS and number of responses by Department, sorted lowest first.

4. Themes: Classify each Comment into exactly one theme from my list (or Unclear) in a new Theme column. Do not change Comment.

5. Sentiment: Add a Sentiment column: Positive, Negative or Mixed for each non-blank Comment.


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LAB 8 · VERIFY

# Turn Engagement Survey Feedback into Recommendations

## ✅ Test it

Promoters 39, passives 61, detractors 60 of 160; eNPS = -13.1. Lowest question Workload Balance (2.71); highest Manager Support (3.76). Human Resources has the lowest department eNPS (-36.4) from only 11 responses, so it is flagged 'interpret with caution'. 134 responses have a comment; after classification Workload is the most frequent theme (about 39 comments - AI labels can vary by a few, which is why the spot-check error rate is recorded). The Recommendations sheet has three rows, each with verified evidence, an owner role, a KPI target and a review date, and Executive_Summary ends with a Share, Revise or Hold decision.


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TOPIC RECAP

# Recap - Use Copilot to Draw Insights and Recommendations

1. LO4 - Draw insights and recommendations: Ask for descriptive evidence before explanations; review Python logic in advanced analysis; classify comments and spot-check them; label OBSERVED versus HYPOTHESIS; and convert verified findings into owned, measurable recommendations.


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WRAP-UP

# Course Summary & Next Steps


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LEARNING OUTCOMES

# What You Achieved

1

## Analyse Data

Explain how Copilot in Excel works (modes, requirements and limits) and use it to analyse, highlight, sort, filter and calculate HR and recruitment data with verified formulas.

2

## Plot Charts

Use Copilot to create, refine and verify charts, PivotCharts and Python visuals that answer specific HR and recruitment questions.

3

## Create Dashboards

Use Copilot to build an interactive, refreshable HR KPI dashboard with PivotTables, KPI cards, charts and slicers, and check every KPI against its definition.

4

## Insights & Recommendations

Use Copilot and Python-based advanced analysis to draw evidence-backed insights and actionable, owned recommendations from workforce and survey data.


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NEXT STEPS

# Continue the Verified Workflow

1. Reuse the sample prompts and the Prompt_Log on an authorised HR or recruitment workbook.
2. Define KPIs in writing before you ask Copilot to build a dashboard.
3. Refresh and re-check dashboards whenever new data arrives.
4. Keep people decisions with accountable people: Copilot informs, it does not decide.


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# Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded C34 course files and use the lab checkpoints to restart safely.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.


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MICROSOFT 365 COPILOT FOR EXCEL · C34

# Thank You!

You can now analyse, chart, dashboard and draw recommendations from HR and recruitment data with Copilot in Excel - while keeping evidence, privacy and human judgement in control.


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